

3.3 EXISTENCE OF SAFETY LAYERS IN ALIGNED LLMs

The cosine similarity adopted in the previous analysis changes non-linearly with respect to angular differences, in this section we calculated the layer-wise angular differences between the two analyses to better highlight the difference between the layers of each aligned LLM when dealing with N-N question pairs and N-M question pairs. Specifically, in r times selection, we calculate the average of the angle gap for each hidden layer, i.e., the average of layer k is calculated as: $\text{avg} \left(\left\{ \angle(v_{n_p}^k, v_{m_q}^k) - \angle(v_{n_p}^k, v_{n_{p'}}^k) \right\}_{p,q,p'} \right)$, where $\angle$ denotes the angle of the two vectors, and avg is the operator of averaging. We show plots of the N-N and N-M pairs layer-wise cosine similarity analysis and angular difference results for several different aligned LLMs in Figure 2.

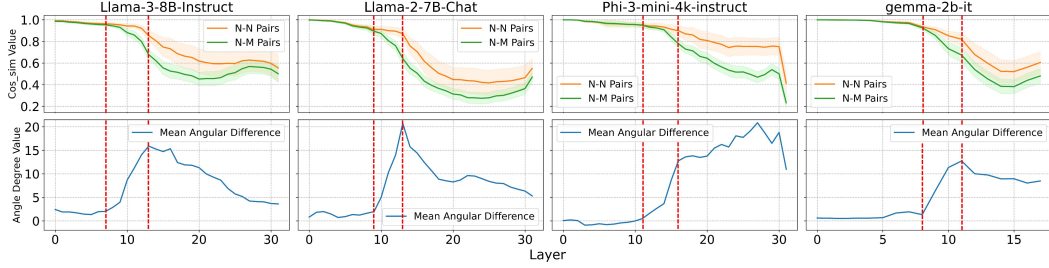


Figure 2: The upper half shows the “Normal-Normal(N-N) Pairs” and “Normal-Malicious(N-M) Pairs” cosine similarity analysis results for each hidden layer of Llama-3-8B-Instruct, Llama-2-7B-Chat, Phi-3-mini-4k-instruct and gemma-2b-it. The lower half displays the mean angular difference between these two cases for each aligned LLM.

It is observed from figure, in the first few layers, the curves show smooth and almost non-existent angle gaps, suggesting that the model processes malicious and normal queries similarly at this stage, indicating no recognition of malicious content in these early layers. However, starting with certain layers in the middle section (the range marked by red dotted lines), the gap between the curve values begins to widen, with an increased growth rate before eventually leveling off. This widening gap reflects the point at which the aligned LLM starts to differentiate between normal and malicious queries, serving as evidence of the existence of safety layers.

Pre-trained LLMs. We also examine the pre-trained versions of aligned LLMs before performing security alignment. Since Phi-3 did not release pre-trained version of their models, we focus on other three pre-trained LLMs. The “N-N Pair” and “N-M Pair” analysis for these pre-trained LLMs are shown in Figure 3. We found that the N-N and N-M curves for each pre-trained LLM do not exhibit any noticeable gaps across all layers. In contrast to aligned LLMs, this absence of a gap aligns with the fact that pre-trained LLMs lack the ability to differentiate between normal and malicious queries. Therefore, the emergence of the safety layers is a direct result of security alignment.

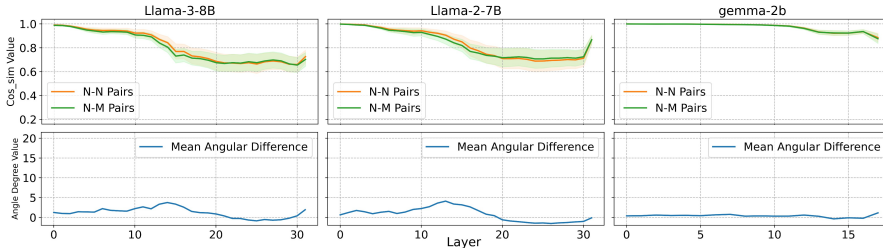


Figure 3: The pre-trained LLMs’ internal layers’ “N-N Pair” and “N-M Pair” analysis.

3.4 LOCALIZATION OF SAFETY LAYERS

In this part, the goal is to locate the most security-related layers for LLM as the safety layers. Although the cosine similarity analysis can recover the existence of the safety layers through the gap in figure 2, locating the safety layers based solely on the range from the appearance of the gap to its increase until the first smoothing is imprecise. This is due to the following reasons: (i) The dimensionality reduction operation of cosine similarity loses part of the information from the hidden layer vectors. (ii) The mean cosine similarity curve is an approximation of the overall tendency of how each aligned LLM treats different normal-normal pairs and normal-malicious pair.

Fortunately, the portion of the curve that grows the fastest from the appearance to the widening of the gap provides a good initial approximate range of the safety layers. With the initial positioning, we further explored: (i) the impact of scaling parameters within certain safety layers on model security, and (ii) the use of the over-rejection phenomenon in aligned LLMs as a clearer indicator of fluctuations in LLM security. These investigations were combined to refine the algorithm for the precise localization of the safety layers.

3.4.1 SCALING PARTIAL SAFETY LAYER PARAMETERS IN INFERENCE

Assuming that the input vector of the i -th layer parameter inside the aligned LLM is h_i , the i -th layer’s output h_{i+1} is formulated as (Meta, 2024):

$$h_{i+1} = \text{FFN}_i(\text{ATTN}_i(h_i) + h_i) + \text{ATTN}_i(h_i) + h_i \quad (1)$$

where $\text{ATTN}_i(h_i)$ and $\text{FFN}_i(h_i)$ represent the outputs of the attention and feed-forward modules of layer i for input h_i , respectively. These modules are the components of each layer of the LLM.

Consistent with the layer vector calculation, each module has a residual connection mechanism during inference. With residual connection, scaling the parameters of a specific layer i by a constant factor α alters the distribution of the output vector. If $\alpha > 1$, the layer’s parameters is amplified, affecting the input to the $i + 1$ -th layer during inference and ultimately changing the final output vector. Essentially, this enhances the layer’s contribution to the autoregressive token generation process. Conversely, if $\alpha < 1$, the layer’s effect is diminished. Scaling the parameters of multiple consecutive layers similarly increases or decreases their collective influence on the final output. However, α should remain close to 1, as excessive deviations can cause confusion within the LLM.

Therefore, when the weights of the parameters in some or all of the layers about safety change, the model’s security performance will also change accordingly. We observed that scaling larger the parameters of the safety layer which is initially localized through cosine similarity analysis can enhance the model’s security, which is reflected in the reduction of the number of queries with propensity to answer in the malicious query dataset (Zou et al., 2023). For example, in the case of Llama-3-8b-Instruct, scaling the parameters of layers 7-12 by a factor of 1.2 reduces the number of queries inclined to generate answers from 29 to 9.

3.4.2 OVER-REJECTION IN SAFETY LAYER PARAMETER SCALING

We define the safety layers as *the most crucial layers for LLM security*. In other words, scaling the targeted safety layers range should have the greatest impact on the security of LLM comparing to other scaled range. However, when using LLM’s response to a malicious dataset as an indicator of how parameter scaling affects its security, a challenge arises: adjusting the upper and lower bounds of the scaled parameter range results in only minimal changes to this indicator because some aligned LLM exhibits strong security performance. For instance, when scaling the parameters by 1.2 times for layer ranges [6,12] and [7,12] of Llama-3-8b-Instruct, the number of queries the LLM is willing to answer from the malicious dataset (Zou et al., 2023) remains consistently low at 9.

To address this, an alternative metric is needed to more effectively reflect the model’s safety performance. Scaling different layer ranges should reveal clearer and more pronounced trends in this new metric, helping to better identify the upper and lower boundaries of the safety layers. Recent studies about the over-rejection phenomenon (Cui et al., 2024; Arditi et al., 2024; Röttger et al., 2023), where aligned LLMs will refuse to answer some non-malicious queries, especially if the queries contains a potentially dangerous verb, bring the new solution to the metric design. As analyzed in Section 3.3, the presence of safety layers in LLMs results from security alignment. Given that over-rejection is a form of “misclassification” arising from enhanced security in alignment, it is directly influenced by these safety layers. Consequently, scaling partial parameters of the safety layers could affect the extent of over-rejection phenomenon.

We further create an over-rejection dataset D_o , each containing queries with potentially dangerous verbs but expressing non-malicious meanings. The number of queries rejected by the LLM in D_o serves as an indicator R_o of security impact. Adjusting the upper and lower bounds of the scaled parameters reveals clear fluctuations in this metric. This is because the over-rejection phenomenon, being an additional effect of security alignment, is more sensitive to changes in the safety layer’s parameters. Therefore, the indicator R_o can serve as a more intuitive measure to help us further determine the upper and lower bounds of the safety layers.

3.4.3 PROGRESSIVE SAFETY LAYERS LOCALIZATION ADJUSTING

With cosine similarity analysis, parameters scaling and the over-rejection dataset D_o , our overall algorithm for precisely localizing the safety layers is as follows:

Step 1. Perform the cosine similarity analysis in section 3.3 for the aligned LLM and locate the initial safety layer, denoted as the range $[i, j]$ from the appearance of the gap to the first smoothing.

Step 2. Use the over-rejection dataset D_o to complete the inference and count the number of queries that the LLM refuses to answer as R_o to evaluate the tested LLM’s baseline degree of over-rejection.

Step 3. By selecting a scaling factor $\alpha > 1$, we up-scale the parameters within layers i to j . We then count the number of problems in dataset D_o that the model refuses to answer, denoted as $R_o^{[i,j]}$. Next, we adjust the upper bound of the safety layers to $j + k$ and measure the over-rejection metric $R_o^{[i,j+k]}$, where k can be negative or positive. There exists a $k = k_u$ such that $R_o^{[i,j+k_u]}$ is greater than $R_o^{[i,j+k_u \pm p]}$ for any p , and we confirm $j + k_u$ as the upper bound. After confirming the upper bound, we perform the same operation to adjust the lower bound of the safety layers, ultimately deriving the range with the largest number of rejected queries as the entire safety layers range $[i - k_l, j + k_u]$.

When N_o is already relatively large, picking an $\alpha < 1$ and performing the same operation but to count the range with the smallest number of rejected queries is also feasible. Due to space limitation, we present how to get the exact value of the hyperparameter α for each LLM in Appendix A.3.4.

Our algorithm locates the safety layers by analyzing the impact of different layer range weight scaling on over-rejection. When we expand the layer range from $[i, j + k - 1]$ to $[i, j + k]$ with $\alpha > 1$, the newly included $j + k$ th layer’s weights are amplified. If this layer is part of the safety layers, the increased influence of security-related parameters leads to a rise in the over-rejection phenomenon. Conversely, if the layer is unrelated or only has little influence to model security, the amplification dilutes the proportion of security parameters, reducing over-rejection. By observing the trend of the over-rejection indicator, we can determine whether the $j + k$ th layer belongs to the safety layers and precisely identify the safety layer’s boundaries. The same principle applies to confirming the lower bound. When scaling with $\alpha < 1$, the model’s security decreases, and the layers most critical to security correspond to points where the over-rejection indicator is minimized.

Furthermore, our localization algorithm does not require the full inference process. In the cosine similarity analysis, we only analyze the hidden layer vectors during the first autoregression step, without needing to generate the entire sequence of tokens. During the layer expansion stage, we focus solely on determining the LLM’s propensity to answer the query. Since aligned LLMs explicitly express this propensity at the start of their response (see Appendix A.3.3 for details), the inference stage requires generating only the first 4 to 8 tokens, making the process highly efficient.

3.5 SAFETY LAYERS OF ALIGNED LLMs

Our safety layer localization method possesses the broad applicability to different aligned LLMs. We show in table 1 the progressive localization process of the safety layers in several aligned LLMs.

		Phi-3-mini-4k-instruct ($\alpha = 0.8, R_o = 270$)					Llama-2-7b-chat ($\alpha = 1.15, R_o = 169$)				
Upper Bound	Scaled Layers range	[11,13]	[11,14]	[11,15]	[11,16]	[11,17]	[9,12]	[9,13]	[9,14]	[9,15]	[9,16]
	Over-Rejection Num	209	190	149	181	189	187	227	237	218	219
Lower Bound	Scaled Layers range	[13,15]	[12,15]	[11,15]	[10,15]	[9,15]	[8,14]	[7,14]	[6,14]	[5,14]	[4,14]
	Over-Rejection Num	237	182	149	177	163	263	268	297	189	202
		Llama-3-8B-Instruct ($\alpha = 1.2, R_o = 139$)					gemma-2b-it ($\alpha = 1.1, R_o = 268$)				
Upper Bound	Scaled Layers range	[7,10]	[7,11]	[7,12]	[7,13]	[7,14]	[8,9]	[8,10]	[8,11]	[8,12]	[8,13]
	Over-Rejection Num	272	241	283	266	256	310	335	368	343	326
Lower Bound	Scaled Layers range	[8,12]	[7,12]	[6,12]	[5,12]	[4,12]	[8,11]	[7,11]	[6,11]	[5,11]	[4,11]
	Over-Rejection Num	334	283	371	358	223	368	371	407	404	323

Table 1: The progressive layer localization process of the four aligned LLMs was conducted using the over-rejection dataset D_o , which consists of 721 problems. Each row in the table presents the range of layers scaled during the adjustment of the upper or lower bounds of the safety layers, along with the corresponding number of problems from the D_o dataset that were refused. The bolded parts indicate the confirmed upper or lower bounds.

As we can see from the trend of the tabular data for each LLM, when $\alpha > 1$, the number of queries that the LLM refuses to answer in the over-reject dataset increases as new layers contributing to security are added to the parameter scaling during the process of confirming the upper or lower bound. Conversely, when a layer that does not contribute to security is included in the parameter scaling, the weight of the safety layers in the overall parameters is relatively reduced, resulting in a decrease in the number of rejected queries. When $\alpha < 1$, the opposite trend occurs (Phi-3).

3.6 DISCUSSION: THREE STAGES OF LLM HIDDEN LAYERS

Through the safety layer localization results above, we observe that the safety layers of the aligned LLM are generally located in the middle part of the model’s parameters, while the very first layers are unrelated to identifying the maliciousness of the input. To further analyze the role of each hidden layer in aligned LLMs during inference, we extract the attention scores for each token from both normal and malicious questions, the heatmaps are shown in figure 4.

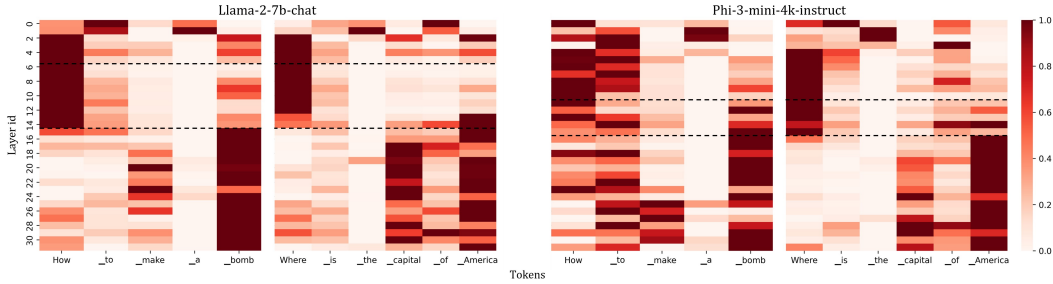


Figure 4: Attention Score Heatmap of Llama-2-7b-chat and Phi-3-mini-4k-instruct. The vertical axis represents each layers, while the horizontal axis corresponds to the input LLM tokens. The darkness of each grid indicates the attention score of a token within a specific layer, reflecting how much attention the layer allocates to that token. Black dashed lines mark the locations of the safety layers, dividing the layers into three distinct sections.

From the figure, we find that in the initial layers before the safety layers, the LLM primarily focuses on syntactic words such as “how” and “the”, without fully grasping sentence semantics. In the layers following the safety layers, the model focuses exclusively on key phrases central to sentence semantics. This pattern is consistent across the LLM’s inference on different queries. More examples with heatmaps are provided in Appendix A.5.2.

Additionally, in Figure 1’s N-N pairs plot, we observe that the fluctuations in cosine similarity across layers driven by different semantic choices in query pairs, only become more pronounced in the layers following the safety layers. Moreover, when we transferred the second half of the hidden layers of Llama-3-8B-Instruct to its homologous pre-train LLM Llama-3-8B, we observed a significant improvement in the model’s ability to answer specific questions logically and semantically, and still without any ability for refusing to answer malicious questions. Conversely, replacing the safety layers and the layers before it did not enhance the pre-trained LLM’s performance (Details are in Appendix A.5.3). These findings all suggest that the layers after the safety layers are closely related to semantic understanding and analysis, indicating a clear segmentation within the model’s layers.

Thus, we propose a three-stage division for the internal layers of aligned LLMs: (i) preliminary sentence confirmation, (ii) detection of malicious intent, and (iii) semantic analysis and understanding. We will continue to explore the roles of these layers in our future work.

4 SAFETY LAYERS: FINE-TUNING JAILBREAK DEFENCE

4.1 SPPFT: SAFELY PARTIAL-PARAMETER FINE-TUNING

The study by Qi et al. (2023) highlights the risk of security degradation in aligned LLMs during full-parameter fine-tuning attack. It reveals that the security of aligned LLMs is reduced regardless of whether the fine-tuning datasets contain harmful data. These findings emphasize the vulnerability of aligned LLMs, presenting a significant challenge for their deployment in real-world applications.

To address this challenge, we propose a safe fine-tuning method based on the safety layers. By freezing the parameters of these safety layers during fine-tuning, their gradients remain unchanged, helping to preserve the security of aligned LLMs. We define this approach as SPPFT (Safely Partial-